

Probability PhD Exam, August 2023

Carefully justify your answers.

1. Let $X, Y : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be random variables. Prove that

$$\mathbb{P}(X + Y \geq 0) \leq \mathbb{P}(X \geq 0) + \mathbb{P}(Y \geq 0).$$

2. This problem has two parts.

1. Give the mathematical definition of:

- (a) convergence almost sure,
- (b) convergence in probability,
- (c) convergence in L^1 ,
- (d) convergence in distribution.

2. For $n \geq 1$, let X_n be uniformly distributed on $\{1, \dots, n\}$, that is

$$\mathbb{P}(X_n = k) = \frac{1}{n}, \quad k \in \{1, \dots, n\}.$$

Prove that $\left\{\frac{X_n}{n}\right\}_{n \geq 1}$ converges to U in distribution, where U is uniform on $[0, 1]$.

3. Let $\{X_i\}_{i \geq 1}$ be a sequence of i.i.d. continuous random variables having a uniform distribution on $[0, 1]$. Define, for $n \geq 1$,

$$Y_n = \max(X_1, \dots, X_n).$$

- 1. Find the cumulative distribution function (CDF) of Y_n , $n \geq 1$.
 - 2. Compute $\mathbb{E}[Y_n]$ and $\text{Var}(Y_n)$.
 - 3. Prove that $\{Y_n\}$ converges to 1 in L^1 . What about almost sure convergence?
4. Let $\{X_i\}_{i \geq 1}$ be a sequence of i.i.d. random variables with a Poisson distribution of parameter λ .
- 1. Let $n \geq 1$. Without justification, what is the distribution of $X_1 + \dots + X_n$?
 - 2. State the weak and strong law of large numbers for i.i.d random variables.
 - 3. Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be an arbitrary bounded continuous function. Prove that

$$\lim_{n \rightarrow +\infty} \sum_{k=0}^{+\infty} e^{-n\lambda} \frac{(n\lambda)^k}{k!} \phi\left(\frac{k}{n}\right) = \phi(\lambda).$$

5. This problem has three parts.

- 1. Give the mathematical definition of standard Brownian motion.
- 2. Let $\{B_t\}$ be a standard Brownian motion. Prove that $\left\{\frac{B_n}{n}\right\}$ converges to 0 almost surely as $n \rightarrow +\infty$ ($n \in \mathbb{N}$).
- 3. Find $\mathbb{P}(B_2 \geq 0, B_1 \leq 0)$.

6. Let $\{M_n\}_{n \geq 0}$ be a sequence of integrable random variables adapted to a filtration $\{\mathcal{F}_n\}$. Assume that for each bounded stopping time T , $\mathbb{E}[M_T] = \mathbb{E}[M_0]$. Show that $\{M_n\}_{n \geq 0}$ is a martingale.
7. Let $\{B_t\}$ be a standard Brownian motion. Define, for $x \in \mathbb{R}$,

$$T_x = \min\{t \geq 0 : B_t = x\}.$$

1. Prove that for all $x \in \mathbb{R}$, T_x is finite almost surely (that is, $\mathbb{P}(T_x < +\infty) = 1$).
2. Find $\mathbb{P}(T_3 < T_{-1})$. You can use a picture as justification.